The Very Standard Tricks in Geometry

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Contents

1 Similar Triangles 2

2 Circles 5

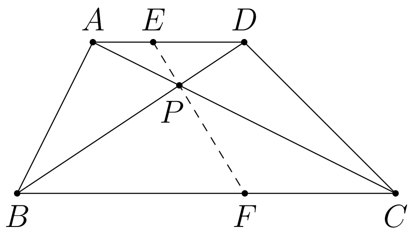
3 Ceva's Theorem and Menelaus' Theorem 10

4 Triangle Geometry 13

1 Similar Triangles

Most geometry problems can be solved using the properties of similar triangles and concyclic points. Therefore, the first thing we do when solving a geometry problem is to figure out pairs of similar triangles and/or groups of concyclic points. Sometimes we need to add extra lines and/or points to deduce useful facts. Drawing accurate figures usually help us to do so.

Problem 1.1. (Intercept theorem(截線定理)) Let $ABCD$ be a trapezoid with $AD//BC$. Points E and F lie on the sides AD and BC respectively. Prove that AC, BD, EF are concurrent if and only if $AE : ED = CF : FB$.



Solution. There are two parts in this question. Firstly, we prove that if AC, BD, EF are concurrent, then $AE : ED = CF : FB$. Note that there are many pairs of similar triangles when there are some parallel lines. Let P be the intersection point of AC, BD, EF. Since $AD//BC$, we have $\triangle APE \sim \triangle CPF$ and $\triangle DPE \sim \triangle BPF$. As we are looking for the lengths AE, ED, CF, FB, we use the similar triangles to deduce $\frac{AE}{CF} = \frac{PE}{PF} = \frac{DE}{BF}$. This is exactly $AE : ED = CF : FB$.

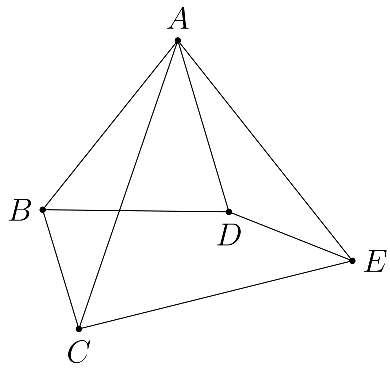
Conversely, assume $AE : ED = CF : FB$. We use a common trick called the method of false position to prove the three lines are concurrent. Suppose the line EP meets BC at F'. It suffices to show that $F' = F$. Indeed, using the above proof, we know that $CF' : F'B = AE : ED$, so that $CF' : F'B = CF : FB$. Now,

$$\begin{aligned}
& \frac{CF'}{F'B} = \frac{CF}{FB} \\
\Rightarrow \quad & \frac{CF'}{F'B} + 1 = \frac{CF}{FB} + 1 \\
\Rightarrow \quad & \frac{CB}{F'B} = \frac{CB}{FB} \\
\Rightarrow \quad & F'B = FB \\
\Rightarrow \quad & F' = F.
\end{aligned}$$

Note that some deductions are only true when F', F lie on the segment BC (instead of lying outside the segment). This completes the proof.

Using a similar proof, we can also deduce that BA, CD, FE are concurrent if and only if $AE : ED = BF : FC$. $\square$

Problem 1.2. Let A, B, C, D, E be five distinct points on the plane such that the rays AC and AD lie inside $\angle BAE$. Prove that $\triangle ABC \sim \triangle ADE$ if and only if $\triangle ABD \sim \triangle ACE$.



Solution. Without loss of generality, assume the ray AC lies between the rays AB and AD.

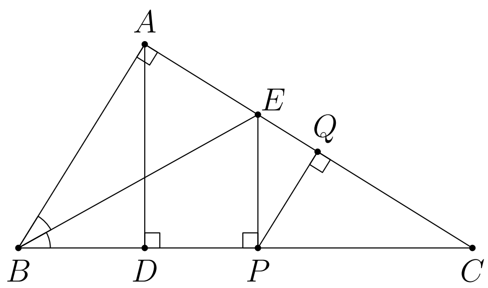
If $\triangle ABC \sim \triangle ADE$, then $\dfrac{AB}{AD} = \dfrac{AC}{AE}$ and $\angle BAC = \angle DAE$. The first equation can be rewritten as $\dfrac{AB}{AC} = \dfrac{AD}{AE}$. Also, we have

$$\angle BAD = \angle BAC + \angle CAD = \angle DAE + \angle CAD = \angle CAE.$$

This implies $\triangle ABD \sim \triangle ACE$. The other direction is similar.

This basic fact of spiral similarity provides a useful way for us to determine similar triangles. Whenever a pair of (directly) similar triangles share a common vertex, we can easily find another pair of similar triangles as suggested. $\square$

Problem 1.3. Let $\triangle ABC$ be a right-angled triangle with $\angle A = 90°$. Let D be the foot of altitude from A. The internal angle bisector of $\angle B$ meets the side AC at E. Let P be the projection of E on BC, and let Q be the projection of P on AC. Prove that $PD = PQ$.



Solution. When there are many right-angles in the figure, usually there are many similar triangles. This gives a lot of relations between the lengths. Also, we can use the angle bisector theorem to get another pair of equal ratio of lengths. These can be combined to yield the desired result.

Note that $\triangle BAC \sim \triangle BDA \sim \triangle ADC \sim \triangle EPC \sim \triangle EQP \sim \triangle PQC$. Among these, the similarity $\triangle ADC \sim \triangle EPC$ tells us some information about DP. For the length PQ, either $\triangle EQP$ or $\triangle PQC$ can be used.

Firstly, since $AD//EP$, we have $\frac{DP}{PC} = \frac{AE}{EC}$ (recall that this is a corollary of the fact $\triangle ADC \sim \triangle EPC$). By the angle bisector theorem, we have $\frac{AE}{EC} = \frac{AB}{BC}$. This suggests we should consider $\triangle BAC$ as well. Indeed, as $\triangle BAC \sim \triangle PQC$, we have $\frac{AB}{BC} = \frac{QP}{PC}$. Now, we have found that

$$\frac{DP}{PC} = \frac{AE}{EC} = \frac{AB}{BC} = \frac{QP}{PC},$$

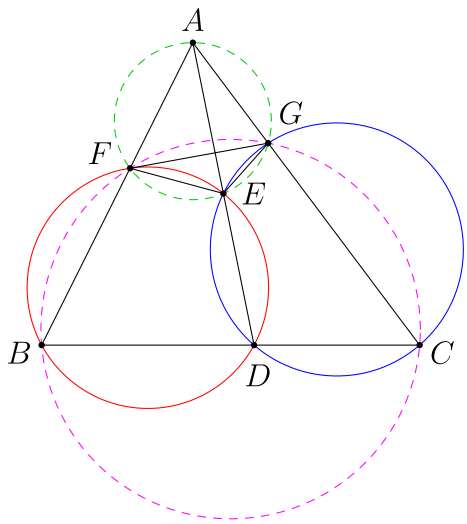
which implies $DP = QP$ as desired. $\square$

The above problem suggests a systematic way for us to find out a proof. Firstly, we write down all pairs (or groups) of similar triangles. Secondly, we use these similar triangles to deduce non-obvious results (which means results that cannot be obtained without proving the triangles are similar, other than those we use to prove the similarities). Sometimes the problem statement gives us hints about which results are important. For harder problems, we may need to use the results further to obtain more similar triangles (or concyclic points) and repeat the same steps.

2 Circles

As mentioned in the previous section, concyclic points always appear in some configurations. Therefore, we have to be familiar with the properties of concyclic points.

Problem 2.1. In $\triangle ABC$, let D be a point on the side BC, and let E be a point on the segment AD. The circumcircle of $\triangle BDE$ meets AB again at F. The circumcircle of $\triangle CDE$ meets AC again at G. Prove that A, F, E, G are concyclic, and B, C, G, F are concyclic.

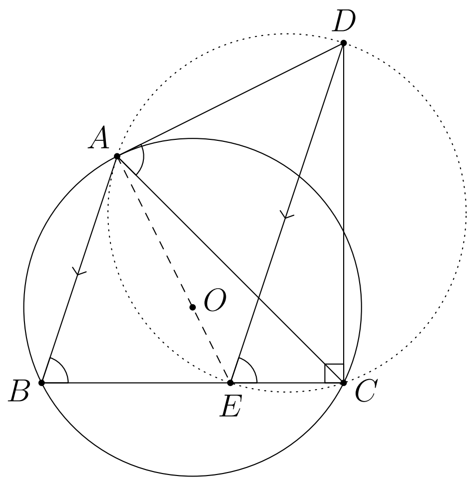


Solution. We first revise some basic properties of concyclic points. Concyclic points are important because there are many angle relations among the angles formed by these points.

Firstly, since B, D, E, F are concyclic, we have $\angle EFA = \angle EDB$. Secondly, since C, D, E, G are concyclic, we have $\angle EDB = \angle EGC$. Combining these, we obtain $\angle EFA = \angle EGC$. This implies A, F, E, G are concyclic. Note that this result still holds if A, E, D are not collinear.

Next, we use another angle relation obtained from concyclic points. Since A, F, E, G are concyclic, we have $\angle AGF = \angle AEF$. Next, since B, D, E, F are concyclic, we have $\angle AEF = \angle DBF$. Combining these, we obtain $\angle AGF = \angle DBF$. This implies B, C, G, F are concyclic. $\square$

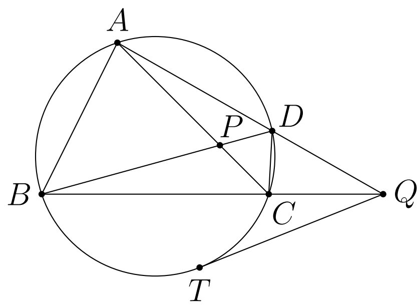
Problem 2.2. Let O be the circumcentre of $\triangle ABC$. The tangent at A to the circumcircle of $\triangle ABC$ meets the line through C perpendicular to BC at D. Let E be the point on BC such that $DE // AB$. Prove that A, O, E are collinear.



Solution. The key step in the following solution is to notice the concyclic points A, E, C, D. If one draws an accurate figure, it is much easier to notice this fact. Alternatively, one may mark some equal angles in the figure and draw some conclusion from these.

Since $AB//DE$, we have $\angle CED = \angle CBA$. As DA is tangent to (ABC), we have $\angle CBA = \angle CAD$. Therefore, we obtain $\angle CED = \angle CAD$. This implies A, E, C, D are concyclic. It follows that $\angle EAD = 180° - \angle DCE = 90°$. Note that we also have $\angle OAD = 90°$ as DA is tangent to (ABC), which has O as the centre. Therefore, A, O, E are collinear. □

One important tool in plane geometry is the power of points (with respect to a circle). As shown in the figure, recall that we have the properties $PA \times PC = PB \times PD$ and $QA \times QD = QB \times QC$.

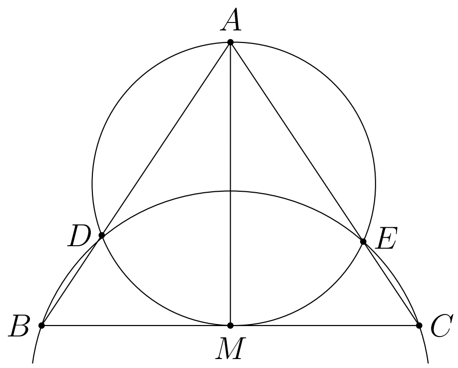


In general, if we draw any line through a point X which cuts a circle at two points Y and Z, then $XY \times XZ$ is a constant. If X lies inside the circle (like the point P), we define the power of X to be $-XY \times XZ$. If X lies outside the circle (like the point Q), we define its power to be $XY \times XZ$. For the latter case, we can consider a special line passing through X, which is the tangent XT to the circle with T being the tangential point. As a limiting case, the power of X is XT^2. Again, the most important fact about powers is that the power of each point with respect to a fixed circle is a constant.

Theorem 2.1. (Power of a point theorem(圓冪定理)) Let O be the centre of a circle, and let R be the radius of the circle. Then the power of any point P with respect to this circle is $OP^2 - R^2$.

Proof. See theorem 5.1 of **Pure Geometry with Directed Angles 2.0**. □

Problem 2.3. (IMO HK TST 2011 Test 2 Problem 1) Given a triangle ABC, let M be the midpoint of BC. The circle passing through A and tangent to BC at M cuts AB and AC at D and E respectively. Suppose B, C, E and D are concyclic, show that $AB = AC$.



Solution. We can use power of points to convert the circle conditions to lengths. Firstly, consider the power of B with respect to $(ADME)$. This gives

$$BD \times BA = BM^2.$$

Similarly, we get

$$CE \times CA = CM^2$$

by considering the power of C with respect to the same circle. As M is the midpoint of BC, the two expressions are equal, so that

$$BD \times BA = CE \times CA. \tag{1}$$

Next, since B, C, E, D are concyclic, we have

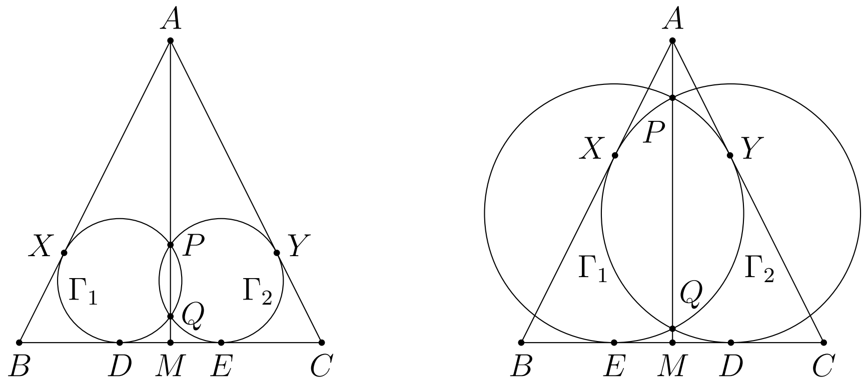
$$AD \times AB = AE \times AC \tag{2}$$

by considering the power of A. Adding (1) and (2), we get

$$AB^2 = AB \times BD + AB \times AD = AC \times CE + AC \times AE = AC^2.$$

This gives $AB = AC$. □

Problem 2.4. (IMO HK TST 2012 Test 2 Problem 2) Let $\triangle ABC$ be a scalene triangle (with no two sides equal). A circle touching sides AB and BC intersects the median AM at P and Q. Another circle touching sides AC and BC also intersects AM at P and Q. Prove that the two circles are the same circle.



Solution. This is another similar problem that can be solved using powers.

Let Γ_1 be the circle touching AB and BC, and let Γ_2 be the circle touching AC and BC. Suppose Γ_1 touches AB and BC at X and D respectively, while Γ_2 touches AC and BC at Y and E respectively.

Suppose on the contrary that $\Gamma_1 \neq \Gamma_2$. Then $D \neq E$ since otherwise both circles pass through three same points. By using powers, we have

$$MD^2 = MP \times MQ = ME^2.$$

This implies M is the midpoint of D and E. As M is also the midpoint of B and C, we have $BD = CE$.

Similarly, we have $AX^2 = AP \times AQ = AY^2$, and so $AX = AY$. It follows that

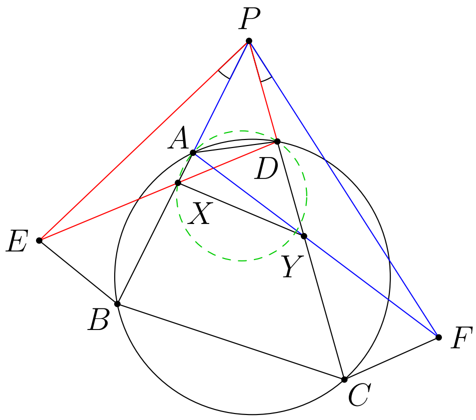
$$AB = AX + BX = AX + BD = AY + CE = AY + CY = AC,$$

contradicting the assumption that $\triangle ABC$ is scalene. Therefore, $\Gamma_1 = \Gamma_2$, which is the incircle of $\triangle ABC$.

Note that it is important that X and Y lie on the sides AB and AC. If exactly one of them lies outside $\triangle ABC$, then the above deduction is incorrect, and in fact it is possible that $\Gamma_1 \neq \Gamma_2$.

In the solution, we have proved some facts about two circles. If two circles intersect at two points P and Q, then for each point on the line PQ, its powers with respect to the two circles are equal (like A and M). Thus, the lengths of tangents from this point to both circles are equal. In particular, PQ bisects the external common tangents (like BC). In fact, the line PQ is called the radical axis of the two circles, and these are some basic properties of the radical axis. $\square$

Problem 2.5. Let $ABCD$ be a cyclic quadrilateral. The extensions of the sides BA and CD intersect at point P. Construct isosceles $\triangle PEB$ and $\triangle PCF$ outside $\triangle PBC$, where $PE = PB$, $PC = PF$ and $\angle EPB = \angle CPF$. If the segments ED and PB meet at point X, the segments FA and PC meet at point Y, prove that XY and BC are parallel.



Solution. In view of Reim's theorem (to be proved below), it suffices to show A, X, Y, D are concyclic.

As $ABCD$ is a cyclic quadrilateral, we have $PA \times PB = PD \times PC$. To use the conditions $PB = PE$ and $PC = PF$, we rewrite this as $PA \times PE = PD \times PF$. This is the same as $\dfrac{PA}{PD} = \dfrac{PF}{PE}$. Also, we have

$$\angle APF = \angle APD + \angle DPF = \angle DPA + \angle APE = \angle DPE.$$

Therefore, we have $\triangle APF \sim \triangle DPE$. It follows that $\angle PAF = \angle PDE$. This implies $\angle XAY = \angle XDY$. Thus, the points A, X, Y, D are concyclic.

Now, since A, B, C, D are concyclic, and A, X, Y, D are concyclic, we can deduce $XY // BC$ by Reim's theorem. The proof is simple. Indeed, we have

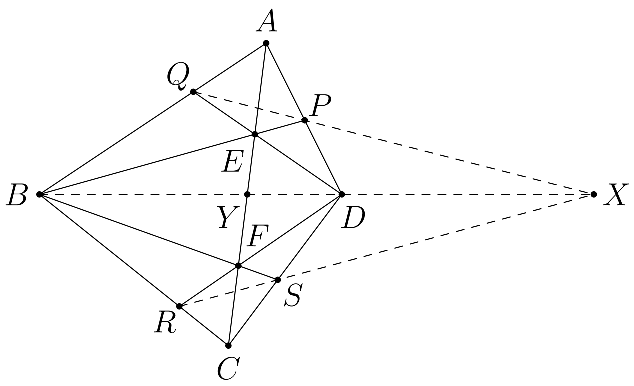
$$\angle PXY = \angle PDA = \angle PBC,$$

which implies $XY // BC$. $\square$

3 Ceva's Theorem and Menelaus' Theorem

Ceva's theorem and Menelaus' theorem are useful when proving concurrent lines and collinear points.

Problem 3.1. (IMO HK TST 2011 Test 1 Problem 4) Let $ABCD$ be a convex quadrilateral. E and F are points on the diagonal AC such that E and F are interior points of triangles ABD and BCD respectively. Suppose BE extend cuts AD at P, DE extend cuts AB at Q, DF extend cuts BC at R and BF extend cuts DC at S. Show that the three lines QP, BD and RS are either parallel or concurrent.



Solution. Let Y be the intersection point of AC and BD. As AY, BP, DQ are concurrent at E, by Ceva's theorem, we have

$$\frac{AQ}{QB} \times \frac{BY}{YD} \times \frac{DP}{PA} = 1. \tag{1}$$

Similarly, we have

$$\frac{CR}{RB} \times \frac{BY}{YD} \times \frac{DS}{SC} = 1.$$

Note that $QP//BD$ if and only if $\dfrac{AQ}{QB} = \dfrac{AP}{PD}$, which is the same as $BY = YD$ by (1). Similarly, $RS//BD$ if and only if $BY = YD$, so that QP, RS are both parallel to BD or not.

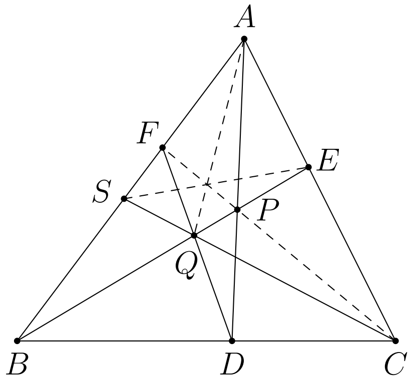
If QP, RS are not parallel to BD, let X_1 be the intersection of QP and BD. By Menelaus' theorem, we have

$$\frac{AQ}{QB} \times \frac{BX_1}{X_1D} \times \frac{DP}{PA} = 1. \tag{2}$$

Comparing (1) and (2), we get $\dfrac{BY}{YD} = \dfrac{BX_1}{X_1D}$. Similarly, letting X_2 be the intersection of RS and BD, we have $\dfrac{BY}{YD} = \dfrac{BX_2}{X_2D}$. This yields $\dfrac{BX_1}{X_1D} = \dfrac{BX_2}{X_2D}$. Since X_1 and X_2 lie outside the segment BD, this implies $X_1 = X_2$. Thus, QP, BD, RS are concurrent.

Therefore, QP, BD, RS are concurrent or parallel. □

Problem 3.2. Let D, E, F be points on the sides BC, CA, AB of $\triangle ABC$ such that AD, BE, CF are concurrent at a point P. The lines DF and BE meet at Q, while CQ and AB meet at S. Prove that AQ, CF, SE are concurrent.



Solution. Using Ceva's theorem, it suffices to prove $\frac{AF}{FS} \times \frac{SQ}{QC} \times \frac{CE}{EA} = 1$. Firstly, we apply Ceva's theorem to the concurrency of AD, BE, CF to get

$$\frac{AF}{FB} \times \frac{BD}{DC} \times \frac{CE}{EA} = 1. \tag{1}$$

Comparing to what we need, to get the term $\frac{SQ}{QC}$, we may, for example, apply Menelaus' theorem on $\triangle SBC$ to get

$$\frac{SQ}{QC} \times \frac{CD}{DB} \times \frac{BF}{FS} = 1. \tag{2}$$

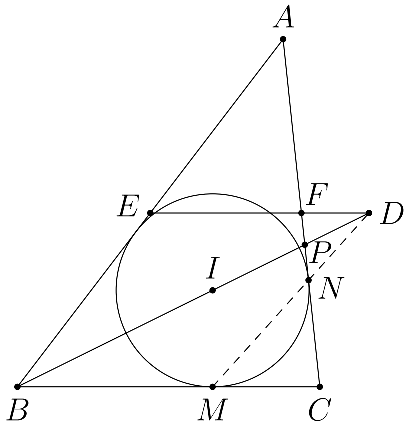
Multiplying (1) and (2), we obtain

$$\frac{AF}{FS} \times \frac{SQ}{QC} \times \frac{CE}{EA} = 1.$$

This proves AQ, CF, SE are concurrent. □

In many cases, we have to use these theorems together with trigonometry and bashing (meaning some tedious algebraic steps). We demonstrate this with one example.

Problem 3.3. (CSMO 2008 Problem 6) As shown in the figure, the incircle $\odot I$ of $\triangle ABC$ touches BC, AC at points M, N respectively. Points E, F are the midpoints of the sides AB, AC respectively. D is the intersection point of the lines EF and BI. Prove that M, N, D are collinear.



Solution. Let a, b, c be the side lengths of BC, CA, AB respectively. Let P be the intersection of BD and AC. In view of the configuration, using Menelaus' theorem, we need to prove $\dfrac{BM}{MC} \times \dfrac{CN}{NP} \times \dfrac{PD}{DB} = 1$. Note that $CM = CN$ as both are tangents from C to $\odot I$. So we just need to prove $\dfrac{BM}{NP} \times \dfrac{PD}{DB} = 1$.

To evaluate $\dfrac{PD}{DB}$, we apply Menelaus' theorem on $\triangle ABP$. This gives

$$\frac{PD}{DB} \times \frac{BE}{EA} \times \frac{AF}{FP} = 1.$$

This is the same as $\dfrac{PD}{DB} = \dfrac{FP}{AF}$ since $BE = EA$. We now need to evaluate NP and FP. Note that BP is the angle bisector. By the angle bisector theorem, we have $\dfrac{PA}{PC} = \dfrac{c}{a}$. Together with $PA + PC = b$, we get $PA = \dfrac{bc}{a+c}$ and $PC = \dfrac{ab}{a+c}$. Also, we have $CN = s - c$ and $CF = \dfrac{b}{2}$. Then we have

$$NP = CP - CN = \frac{ab}{a+c} - (s-c) = \frac{c^2 - a^2 + ab - bc}{2(a+c)} = \frac{(c-a)(c+a-b)}{2(a+c)}$$

and

$$FP = CF - CP = \frac{b}{2} - \frac{ab}{a+c} = \frac{bc - ab}{2(a+c)} = \frac{b(c-a)}{2(a+c)}.$$

Therefore, we find that

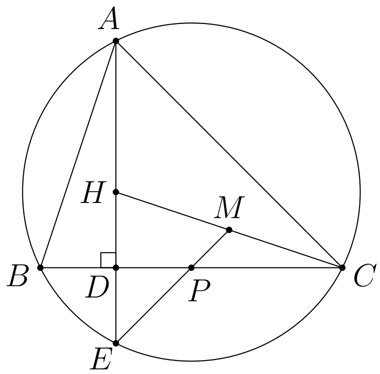
$$\frac{BM}{NP} \times \frac{PD}{DB} = \frac{BM}{NP} \times \frac{FP}{AF} = \frac{s-b}{\dfrac{(c-a)(c+a-b)}{2(a+c)}} \times \frac{\dfrac{b(c-a)}{2(a+c)}}{\dfrac{b}{2}} = \frac{a+c}{c-a} \times \frac{c-a}{a+c} = 1.$$

This completes the proof. $\square$

4 Triangle Geometry

To solve problems in triangle geometry, one needs to memorize a lot of facts about the triangle centres. Here we are only going to go through several problems, but these are not enough to cover all important properties of the triangle centres. One should revise more and do more to get familiar with the facts.

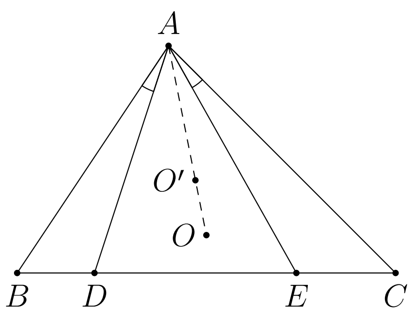
Problem 4.1. Let $\triangle ABC$ be an acute triangle. H is the orthocentre of $\triangle ABC$. The line AH meets BC at D and meet the circumcircle of $\triangle ABC$ again at E. Let M be the midpoint of CH. The line EM and BC meet at P. Prove that $DP : PC = 1 : 2$.



Solution. The $1:2$ ratio suggests using a property of the centroid. In other words, the result holds if we can prove that P is the centroid of $\triangle CEH$. Noting that EM is a median of this triangle, this guess should be correct.

Indeed, recall that E is the reflection of H in the line BC. Therefore, D is the midpoint of HE. Since M is the midpoint of HC, the intersection point P of the medians CD and EM is the centroid of $\triangle HEC$. This proves $DP : PC = 1 : 2$. □

Problem 4.2. In $\triangle ABC$, let D and E be two points on the side BC such that $\angle BAD = \angle CAE$. Let O be the circumcentre of $\triangle ABC$ and let O' be the circumcentre of $\triangle ADE$. Prove that A, O, O' are collinear.



Solution. A simple angle chasing can solve this problem. This is a simple practice on the relation of angles related to the circumcentre.

For convenience, we only consider the configuration as shown since other cases are similar. It suffices to show that $\angle DAO' = \angle DAO$. Indeed, we have

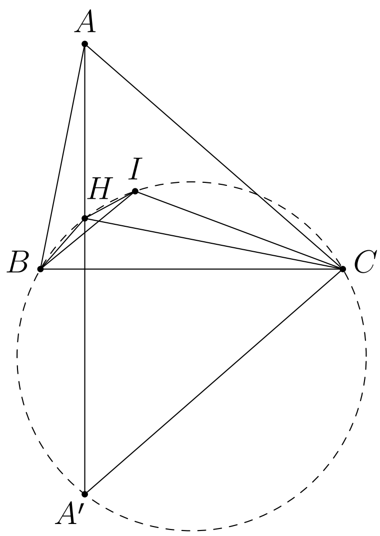
$$\angle DAO' = \frac{180^\circ - \angle AO'D}{2} = 90^\circ - \angle AED = 90^\circ - \angle EAC - \angle ACE$$

and

$$\angle DAO = \angle BAO - \angle BAD = \frac{180^\circ - \angle AOB}{2} - \angle BAD = 90^\circ - \angle ACB - \angle BAD.$$

The two expressions are equal since $\angle BAD = \angle EAC$. The result then follows. $\square$

Problem 4.3. In $\triangle ABC$, $\angle BAC = 60^\circ$ and $AB < AC$. Let H and I be the orthocentre and incentre of $\triangle ABC$ respectively. Prove that $\angle AHI = \frac{3}{2}\angle ACB$.



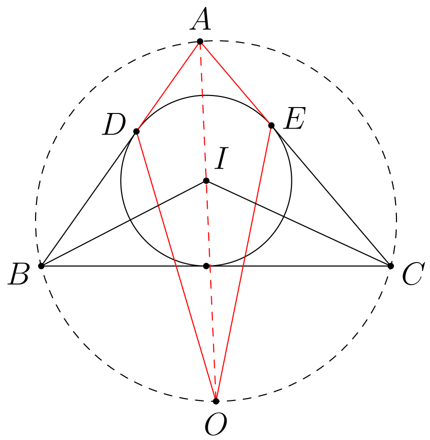
Solution. Let $\angle ACB = 2x$. Then $\angle ACI = \angle ICB = x$. Note that we need to prove $\angle AHI = 3x$. To prove this geometrically, we try to construct an angle which equals $3x$ and compare this with $\angle AHI$. Indeed, reflecting A in the line BC to the point A', we find that

$$\angle ICA' = \angle ICB + \angle BCA' = x + \angle ACB = x + 2x = 3x.$$

To show that $\angle AHI = \angle ICA'$, it is the same as proving A', C, I, H are concyclic. This is a standard result for a triangle with $\angle A = 60^\circ$.

Firstly, note that $\angle BHC = 180^\circ - \angle BAC = 120^\circ = 180^\circ - \angle BA'C$. Thus, B, A', C, H are concyclic. Secondly, as $\angle BIC = 90^\circ + \frac{1}{2}\angle BAC = 120^\circ = \angle BHC$, the point I also lies on (BCH). Therefore, B, A', C, I, H are concyclic, and so $\angle AHI = \angle ICA' = 3x$ as desired. $\square$

Problem 4.4. (CGMO 2012 Problem 5) The incircle $\odot I$ of $\triangle ABC$ is tangent to the sides AB, AC at points D, E respectively. The circumcentre of $\triangle BCI$ is O. Prove that $\angle ODB = \angle OEC$.



Solution. Here we use an important property about the incentre. Recall that O lies on the internal angle bisector of $\angle BAC$ (and it lies on the circumcircle of $\triangle ABC$). By $\angle OAD = \angle OAE$, $AD = AE$ and the common side AO, we have $\triangle OAD \cong \triangle OAE$. Therefore, we find that $\angle ODA = \angle OEA$. Thus,

$$\angle ODB = 180^\circ - \angle ODA = 180^\circ - \angle OEA = \angle OEC.$$

$\square$